Sabee Grewal

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Links: *Personal Website*, *Google Scholar Profile*.

Education

2021–Present **Ph.D. in Computer Science** *University of Texas at Austin.*
Advisor: Scott Aaronson

2021 **M.S. in Electrical and Computer Engineering** *University of Texas at Austin.*

2019 **B.S. in Physics** *University of Texas at Austin.*

2017 **B.S. in Electrical and Computer Engineering** *University of Texas at Austin.*

Publications

Author ordering in my field is alphabetical by default (all exceptions below are evident).

- S. Aaronson, M. Bavarian, T. Cubitt, **S. Grewal**, G. Gueltrini, R. O'Donnell, M. Raat. Computability Theory of Closed Timelike Curves. *In Submission.* [ARXIV]
- **S. Grewal**, V. M. Kumar. Improved Circuit Lower Bounds and Quantum-Classical Separations. *In Submission.* [ARXIV] [ECCC]
- **S. Grewal**, V. Iyer, W. Kretschmer, D. Liang. Agnostic Tomography of Stabilizer Product States. *In Submission.* [ARXIV]
- **S. Grewal**, V. Iyer, W. Kretschmer, D. Liang. Pseudoentanglement Ain't Cheap. *Presented at TQC 2024. In Submission.* [ARXIV]
- S. Aaronson, **S. Grewal**, V. Iyer, S. C. Marshall, R. Ramachandran. $\mathsf{PDQMA} = \mathsf{DQMA} = \mathsf{NEXP}$: QMA With Hidden Variables and Non-collapsing Measurements. *In Submission.* [ARXIV]
- **S. Grewal**, J. Yirka. The Entangled Quantum Polynomial Hierarchy Collapses. *39th Computational Complexity Conference (CCC 2024). Leibniz International Proceedings in Informatics (LIPIcs), Volume 300, pp. 6:1-6:23, Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2024).* [ARXIV] [ECCC] [DOI]
- **S. Grewal**, V. Iyer, W. Kretschmer, D. Liang. Efficient Learning of Quantum States Prepared With Few Non-Clifford Gates. *Presented at QIP 2024. In Submission.* [ARXIV]
- **S. Grewal**, V. Iyer, W. Kretschmer, D. Liang. Improved Stabilizer Estimation via Bell Difference Sampling. *56th Annual ACM Symposium on Theory of Computing (STOC 2024). Presented at QIP 2024.* [ARXIV] [DOI]
- S. Aaronson, **S. Grewal**. Efficient Tomography of Non-Interacting-Fermion States. *18th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2023), Leibniz International Proceedings in Informatics (LIPIcs)* 266, 12:1–12:18 (2023). [ARXIV] [DOI]
- **S. Grewal**, V. Iyer, W. Kretschmer, D. Liang. Low-Stabilizer-Complexity Quantum States Are Not Pseudorandom. *14th Innovations in Theoretical Computer Science Conference (ITCS 2023), Leibniz International Proceedings in Informatics (LIPIcs)* 251, 64:1–64:20 (2023). **ITCS 2023 Best Student Paper.** [ARXIV] [DOI]
- D. E. Koh, **S. Grewal**. Classical Shadows With Noise. *Quantum:* 6, 776 (2022). [ARXIV] [DOI]

Selected Awards

2023 **Best Student Paper Award** *14th Innovations in Theoretical Computer Science (ITCS 2023)* .

2020 **Teaching Excellence Award** *Cockrell School of Engineering*, UT Austin.

2015 **Teaching Excellence Award** *Cockrell School of Engineering*, UT Austin.

2014 **Teaching Excellence Award** *Cockrell School of Engineering*, UT Austin.

2014 **Distinguished College Scholar**, UT Austin, (for being in the top 4% of the college).

Experience

Research Experience

Summer 2024 **Quantum Research Scientist Intern** *IBM*, Yorktown Heights, NY.

Spring 2024 **Long-Term Visitor** *Simons Institute for the Theory of Computing*, Berkeley, CA.

2021–present **Graduate Research Assistant** *Quantum Information Center*, UT Austin.

summer 2020 **Quantum Computing Research Intern** *Zapata Computing*, Boston, MA.

summer 2019 **Quantum Computing Research Intern** *Zapata Computing*, Boston, MA.

Engineering Experience

fall 2020 **Senior Software Engineer** *Zapata Computing*, Boston, MA.

2017–2019 **Software Engineer II** *Microsoft Corporation*, Redmond, WA.

summer 2016 **Software Engineer Intern** *Microsoft Corporation*, Redmond, WA.

summer 2015 **Software Engineer Intern** *Microsoft Corporation*, Redmond, WA.

summer 2014 **CPU Design Intern** *Intel Corporation*, Austin, TX.

Selected Talks & Presentations

2024 **Improved Stabilizer Estimation via Bell Difference Sampling**.
UT Austin CS Theory Seminar

2024 **The Entangled Quantum Polynomial Hierarchy Collapses**.
Computational Complexity Conference (CCC) 2024

2024 **Improved Stabilizer Estimation via Bell Difference Sampling**.
56th ACM Symposium on Theory of Computing (STOC 2024)

2024 **Learning Quantum States With Respect To The Stabilizer Formalsim**.
27th Conference on Quantum Information Processing (QIP 2024)

2023 **Efficent Tomography of Non-Interacting-Fermion States**.
18th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2023)

2023 **Low-Stabilizer-Complexity Quantum States Are Not Pseudorandom**.
14th Innovations in Theoretical Computer Science (ITCS 2023)

Teaching

Fall 2023 **Teaching Assistant** *Quantum Information Science I*, UT Austin.

Fall 2021 **Head Teaching Assistant** *Introduction to Computing Systems*, UT Austin.

Spring 2021 **Teaching Assistant** *Combinatorics and Graph Theory*, UT Austin.

Fall 2019 **Head Teaching Assistant** *Introduction to Computing Systems*, UT Austin.

Spring 2019	**Teaching Assistant** *Software Testing*, UT Austin.
Fall 2016	**Teaching Assistant** *Engineering Programming Languages*, UT Austin.
Fall 2015	**Teaching Assistant** *Introduction to Computing Systems*, UT Austin.
Fall 2014	**Teaching Assistant** *Software Design and Implementation*, UT Austin.

Skills

Programming Languages: *Python, Java, Scala, C++, Bash, LaTeX, Git, AWS, Azure*.

Professional Service

Conference Reviewer: *QIP 2023, QIP 2024, QCTIP 2024, TQC 2024, ICALP 2024, AQIS 2024, YQIS 2024, ITCS 2025, QIP 2025, STOC 2025*.

Journal Reviewer: *Theory of Computing Systems (2024), PRX Quantum (2024)*.

Last updated: November 2024